

01 / Three harnesses. Three box wires.

POOLDose

REV 13

Ferrule branches at existing devices · all 120 V wires are 14 AWG · small AC terminals each receive one blue ferrule

A AC supply / B Switching / C Low voltage · device order matches the enclosure photo

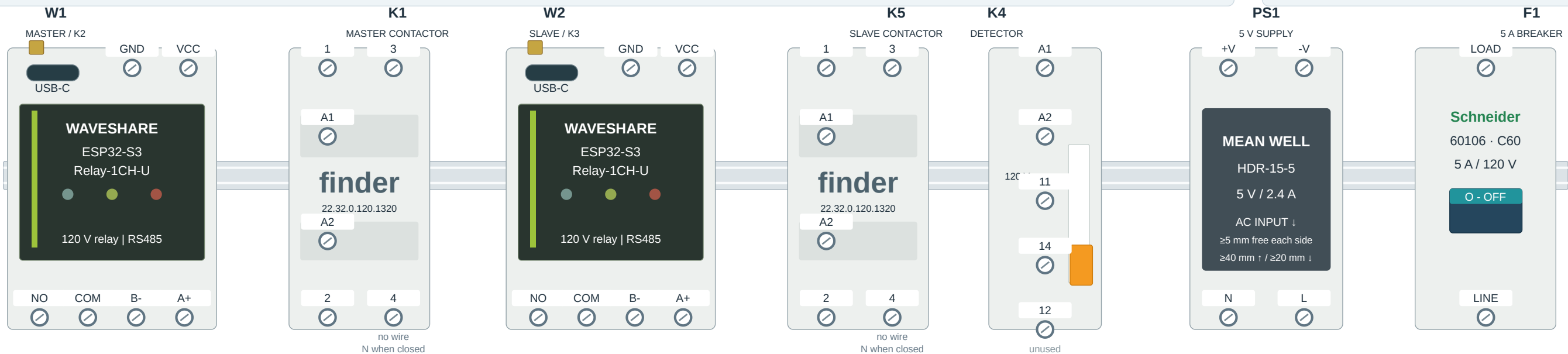
SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification

Ferrules and checkout

Three LARGE RED ferrules: F1 LOAD, K5-3 and K1-3. One YELLOW: K5-2. Remaining 14 AWG device ends: BLUE.
Check: supply output 5.00 V with all relays closed; communication, timed run, STOP and flow-loss shutdown.
Calibrate with water before chemical service (docs/pump-calibration.md). Assembly and checkout: docs/build-instructions.html.
WARNING: K5-3 and K1-3 exceed the Finder terminal rating. Reference build only; do not copy. See docs/wiring-schedule.md.

Finder operating power

A1/A2: 120 V AC coil supply.
1-2 and 3-4: NO contact pairs.
LED + mechanical indicator.
Wiring scope: see end notes.



A — AC SUPPLY

Black hot + separate white neutral.
Larger screws carry the branch ferrules.
Sheets 02–03.

B — SWITCHING

Red coil leads + outlet return.
K5-2 shares B04 + B05.
Sheet 04.

C — LOW VOLTAGE

5 V, RS485 + GPIO tails.
PS1 → W2 → W1 power chains.
Sheets 05–07.

Only these leads cross the metal-box boundary

A01 BLACK / 14 AWG OUT → F1 bottom LINE. A02 leaves the top LOAD terminal.
A08 WHITE / 14 AWG OUT → K5 POWER terminal 3. Both whip neutrals join inside the box.
B04 RED / 14 AWG IN ← final switched output. Wago → outlet-whip BLACK.
Two 3-wire whips end in the box; all whip and 120 V harness wires are stranded 14 AWG.
Whip grounds are joined and bonded to the metal box.

METAL JUNCTION BOX

Wagos join the source/outlet whips to three control leads.

A01 · BLACK OUT → F1 bottom LINE

A08 · WHITE OUT → controls

B04 · RED IN → outlet-whip BLACK

Neutral → outlet whip

Ground → outlet whip
PE → metal-box bond

Neutral and PE are separate nets. The green reference stays INSIDE the metal box; it is not a harness lead.

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

6 requested shared ferrules

02 / Harness A / BLACK hot branches

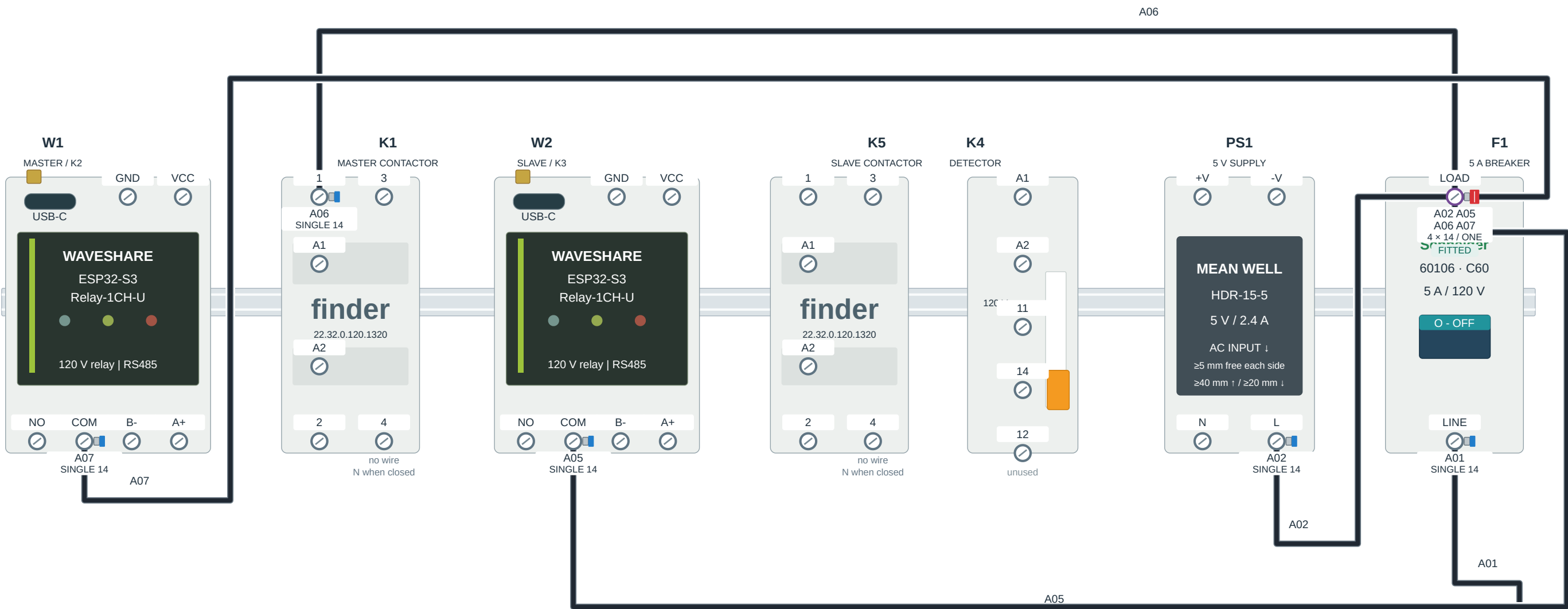
POOLDose

REV 13

F1 top LOAD: one ferrule / four 14 AWG wires → PS1 L, W2 COM, K1 terminal 1 and W1 COM

BLACK = stranded 14 AWG · every destination is SINGLE BLUE · Waveshare COM is the relay contact, never VCC

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification



Four individual leads leave ONE ferrule at F1 LOAD

A02 → Mean Well L. A05 → Slave W2 COM.
A06 → Master Finder K1 terminal 1. A07 → Master W1 COM.
Each far end is one 14 AWG wire in one blue ferrule.
A01 is the single incoming black lead at bottom LINE.

Branch ferrule

One LARGE RED ferrule at F1 top LOAD.
A02/A05/A06/A07: four 14 AWG wires in that one end.
Tighten and check each wire for retention.
Kit chart: 8 AWG ferrule label / 12 mm barrel / 4.9 mm OD.

METAL BOX

N OUT J1 IN L OUT
A01 BOX

Wagos: source + outlet whips
Whips: stranded 14 AWG
N and PE remain separate

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

This sheet: 1 shared ferrule

03 / Harness A / WHITE neutral branches

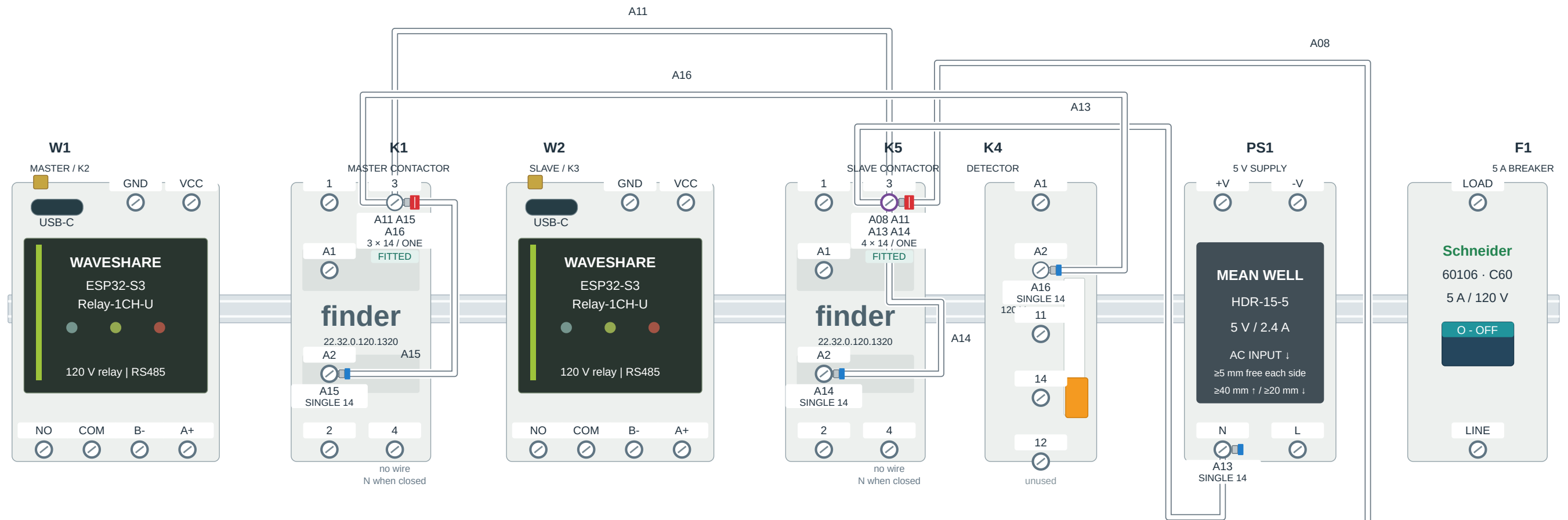
BOX N → Finder K5 terminal 3 → Finder K1 terminal 3 · separate single-blue leads to all four neutral loads

POOLDose

REV 13

WHITE = stranded 14 AWG · branch at POWER terminal 3 · all A1/A2 and Mean Well N ends are SINGLE BLUE

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification



K1 POWER terminal 3 / three wires

A11 IN + A15 to K1 A2 + A16 to K4 A2.
Over Finder rating: reference build only.
K1 A2 gets only A15 in one blue ferrule.

K5 POWER terminal 3 / four wires

A08 IN + A11 bridge + A13 + A14.
A13 → PS1 N. A14 → K5 A2; each single blue.
Over Finder rating: reference build only.

No neutral path crosses a contact

A11 is a real wire between the two terminal 3s.
Both terminal 4s have no external wire.
A closed 3–4 contact only makes its empty 4 neutral.

METAL BOX

N OUT J1 IN L OUT

A08 BOX

Wagos: source + outlet whips
Whips: stranded 14 AWG
N and PE remain separate

A1/A2 are coil terminals, distinct from 1/2/3/4. Both coils retain neutral in every contactor state.

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

This sheet: 2 shared ferrules

04 / Harness B / switched wires

POOLDose

REV 13

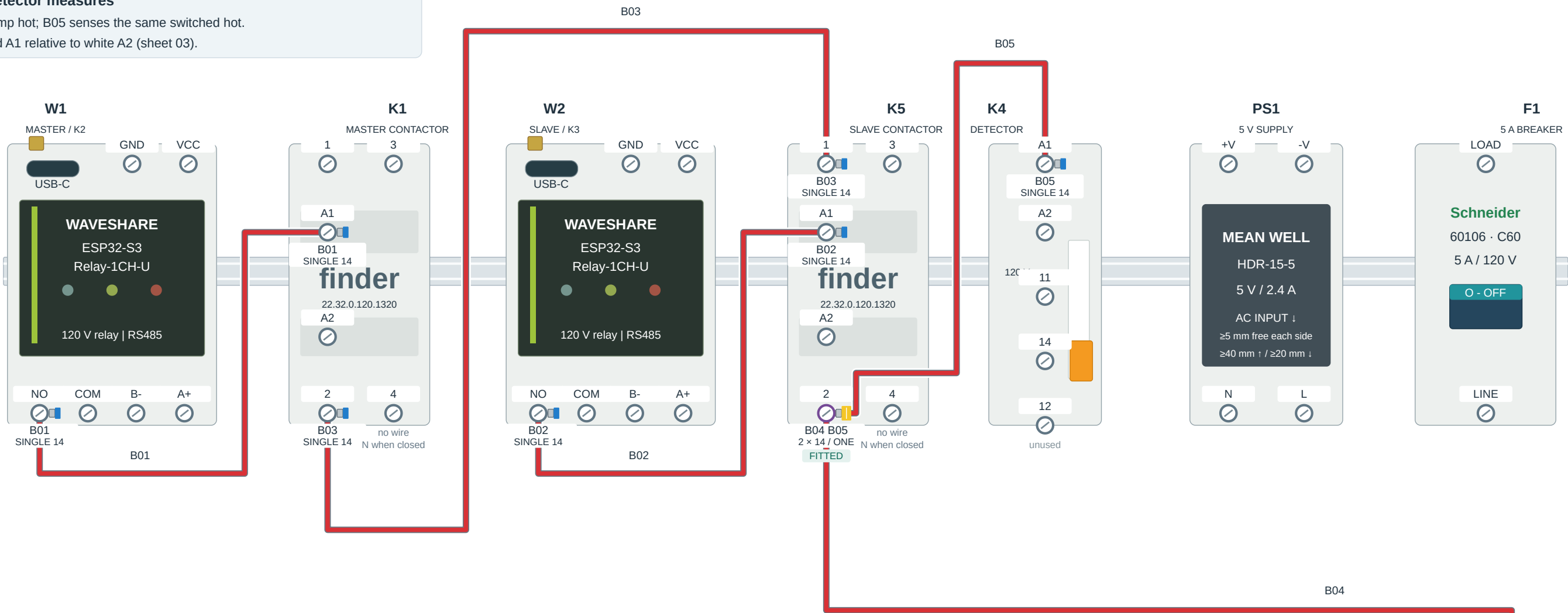
B01–B05 · all RED 14 AWG · independent coil jumpers and series pump-power path

B01 = Master → K1 coil · B02 = Slave → K5 coil · B03 = series power link · B04/B05 = final output + detector

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification

What the detector measures

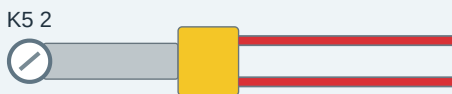
B04 feeds pump hot; B05 senses the same switched hot.
K4 senses red A1 relative to white A2 (sheet 03).



Finder coil power: A1/A2 are distinct from 1/2

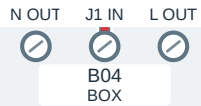
B01: W1 NO → K1 A1. B02: W2 NO → K5 A1.
Both A2 terminals receive white neutral from harness A.
1-2 switches pump hot. Terminal 3 is neutral; 4 has no wire.
The LED follows coil power; indicators need no extra wiring.

Both red wires share ONE ferrule at Finder K5 terminal 2



Detector A1 receives B05 alone: SINGLE 14.
Its A1 and A2 each receive one 14 AWG wire in one ferrule.
K5-2: one YELLOW ferrule holds both red 14 AWG wires.

METAL BOX



Wagos: source + outlet whips
Whips: stranded 14 AWG
N and PE remain separate

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

This sheet: 1 shared ferrule

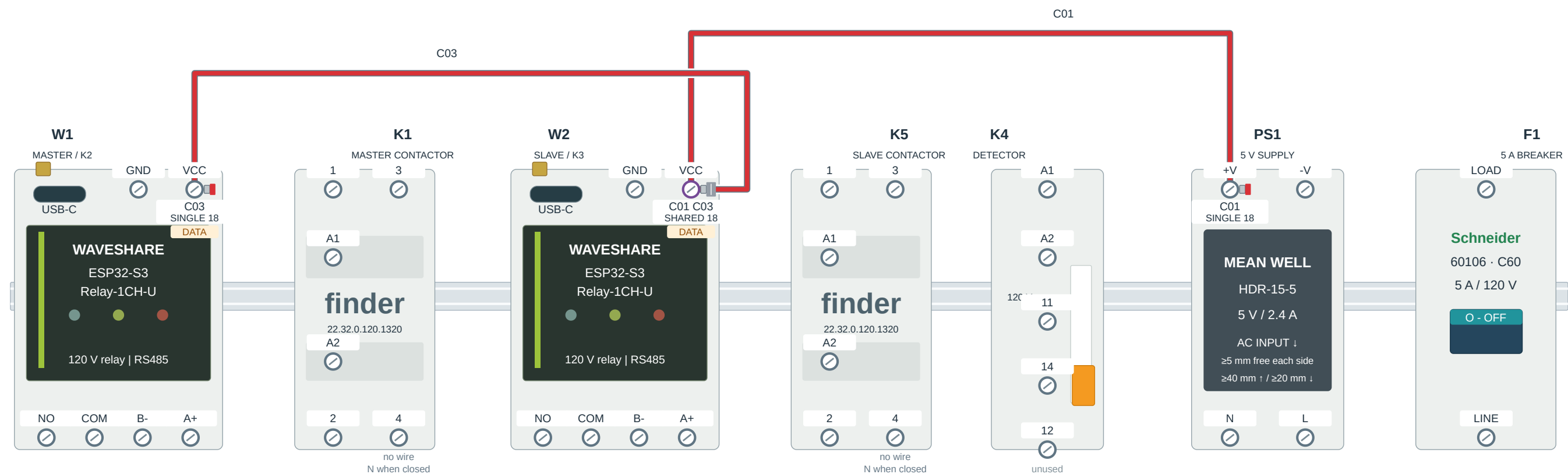
05 / Harness C / RED +5 V daisy chain

POOLDose

REV 13

5 V DC ONLY · C01 arrives and C03 continues from the SAME W2 VCC screw

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification



The red low-voltage harness

C01: PS1 +V → W2 VCC.

C03: same W2 VCC → W1 VCC.

W2 VCC: C01 + C03 together in ONE 2 × 18 AWG twin.

PS1 +V and W1 VCC each have one single-ferruled wire.

Existing terminal checks

DATA: record Waveshare single/twin wire ranges, strip and torque.

PS1 stays single on +V; confirm the actual ferrule/barrel.

Use the top VCC screw, not GPIO 3V3.

The requested shared end is visible at W2; no added connector is drawn.

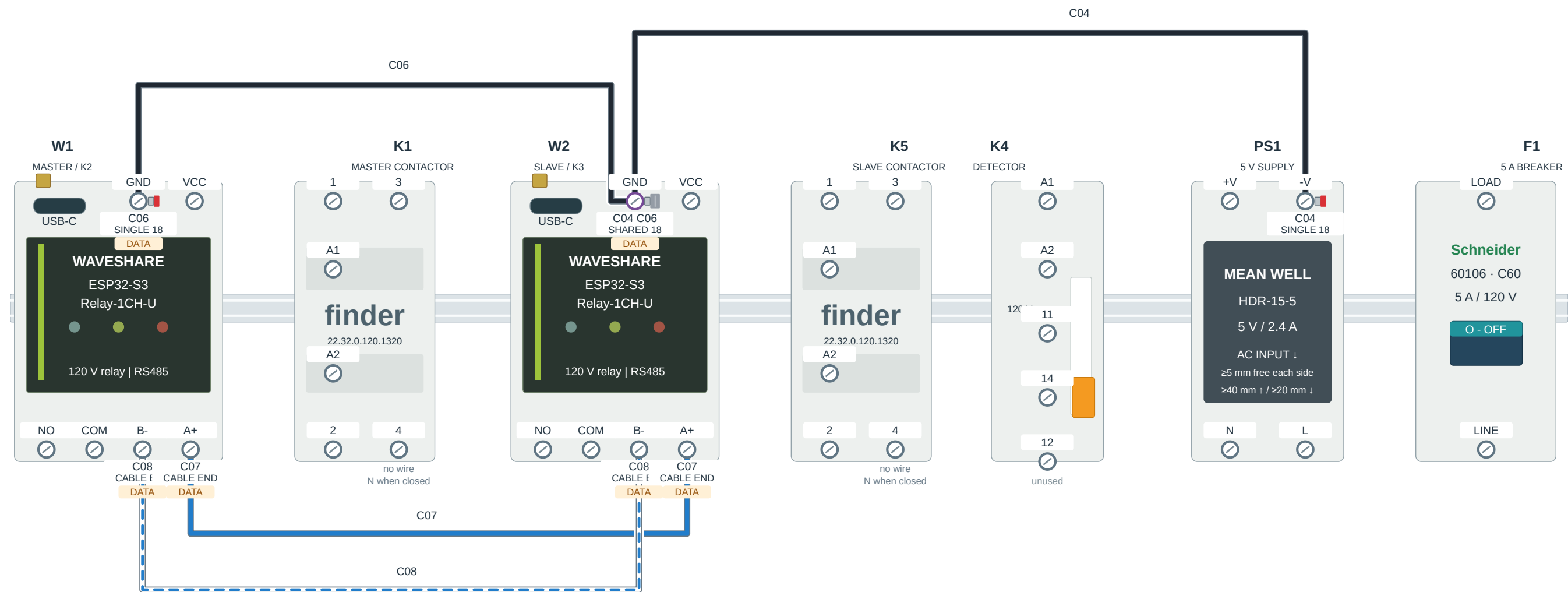
06 / Harness C / BLACK 0 V daisy chain + RS485

POOLDose

REV 13

BLACK = DC 0 V, separate from AC neutral and PE · BLUE = A+ · WHITE/BLUE = B-

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification



Intact Cat6A cable

C07 blue: A+ ↔ A+. C08 white/blue: B- ↔ B-.
Preserve the cable jacket and pair twist.
Enable both 120 Ω jumpers.
Check cable gauge/construction; solid conductors stay bare.

The black low-voltage harness

C04: PS1 -V → W2 GND. C06: same W2 GND → W1 GND.
W2 GND: C04 + C06 together in ONE 2 × 18 AWG twin.
PS1 -V and final W1 GND each have one single-ferruled wire.
DATA: record the WaveShare terminal/ferrule specification.

07 / Harness C / GPIO plugs and pull-up

POOLDose

REV 13

Photographed local pull-up splice · ORANGE detector signal / BLACK return · completed RED/GREEN telephone-cord flow tail

J5 plug colors: 1 YELLOW / GPIO1 · 2 ORANGE / GPIO2 · 3 RED / +3.3 V · 4 BLACK / GND

Slave: RED → R1 → one splice with YELLOW GPIO1 + ORANGE C13. BLACK → BLACK C14. Insulate each joint separately.

C13 alone → detector 14; C14 alone → detector 11. Both are single-wire ends. The thin ORANGE GPIO2 lead stays unused.

W2 / SLAVE · four-wire J5 plug

Photo layout · orange signal / black return

1 · GPIO1 · YELLOW

2 · GPIO2 · ORANGE

3 · +3.3 V · RED

4 · GND · BLACK

YELLOW GPIO1 joins the resistor/output splice at JS2

Cap this ORANGE plug wire individually. GPIO2 has no connection.

R1 · 270 Ω · ¼ W

JR1
RED lead joined directly to R1
Either resistor lead may face 3.3 V.
C12 is the resistor's own lead; sleeve it.

JS2: YELLOW GPIO1 + R1 lead C12 + ORANGE C13 in one local splice.

K4-14 and K4-11 each receive ONE 18 AWG wire in a SINGLE ferrule.

Insulate JS2, the RED/R1 splice and the BLACK splice separately; support the assembly.
C14 · BLACK · to K4-11

JS3
C14
SOLDER

K4 / DETECTOR

Contact terminals only

11
C14
SINGLE 18

11 = common
14 = normally open

11–14 closes when
the outlet is powered.

14
C13
SINGLE 18

12
unused

Unplug GPIO plugs for soldering and resistance checks. Slave pins 3 ↔ 1: 270 Ω ±5%; pins 1 ↔ 4: open with K4 released.

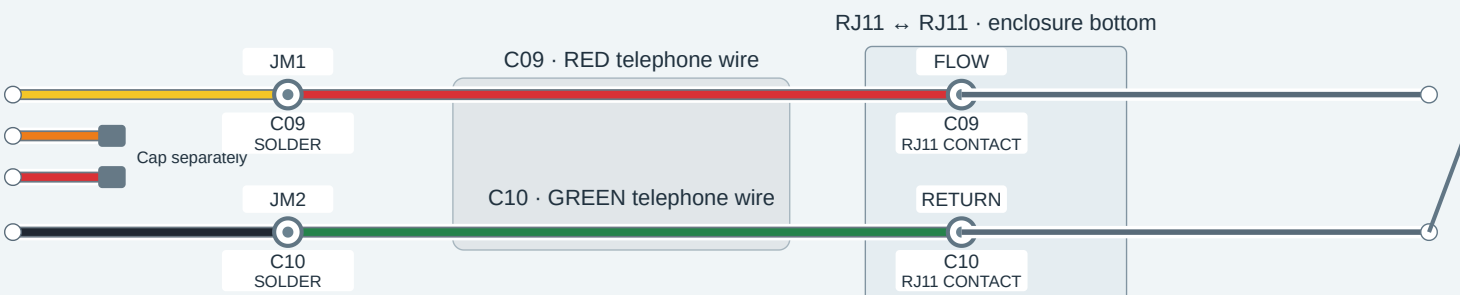
W1 / MASTER · completed flow tail

1 · GPIO1 · YELLOW

2 · GPIO2 · ORANGE

3 · +3.3 V · RED

4 · GND · BLACK



Photos: YELLOW GPIO1 → RED cord wire; BLACK GND → GREEN cord wire. Master tail is shown finished in heat-shrink.

S1 · generic Hayward-style flow switch

RJ11 plug connects to the other side.

Keep the installed contact mapping.

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

This sheet: 0 shared ferrules

08 / One ferrule / multiple conductors

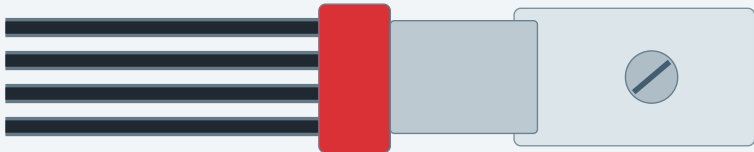
Six shared device locations · K4 detector terminals 11 and 14 now each receive a SINGLE 18 AWG wire

POOLDose

REV 13

F1 top LOAD / 4 × 14 AWG

A02 + A05 + A06 + A07



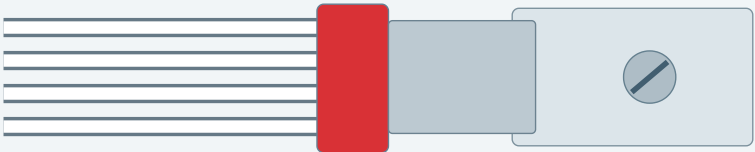
ONE crimped end / 4 separate conductors

LARGE RED / 12 mm kit barrel.

Single blue at each small-terminal destination.

K5 POWER terminal 3 / 4 × 14 AWG

A08 + A11 + A13 + A14



ONE crimped end / 4 separate conductors

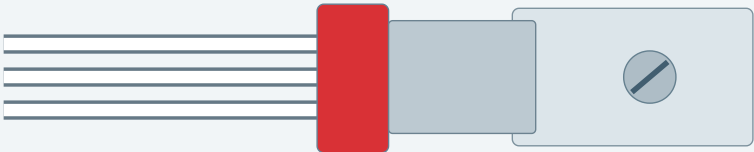
LARGE RED / 12 mm kit barrel.

Single blue at each small-terminal destination.

Exceeds Finder rating: reference build only.

K1 POWER terminal 3 / 3 × 14 AWG

A11 + A15 + A16



ONE crimped end / 3 separate conductors

LARGE RED / 12 mm kit barrel.

Single blue at each small-terminal destination.

Exceeds Finder rating: reference build only.

EXISTING SCREW	ALL WIRES IN ONE FERRULE	INSULATION / AWG	TERMINAL NOTE
F1 LOAD	A02 + A05 + A06 + A07	BLACK · 4 × 14	FITTED — LARGE RED; tightened and connected as drawn.
K5 terminal 3	A08 + A11 + A13 + A14	WHITE · 4 × 14	FITTED — LARGE RED; exceeds Finder rating, reference build only.
K1 terminal 3	A11 + A15 + A16	WHITE · 3 × 14	FITTED — LARGE RED; exceeds Finder rating, reference build only.
K5 terminal 2	B04 + B05	RED · 2 × 14	FITTED — YELLOW; two 14 AWG wires in the one kit ferrule.
W2 VCC	C01 + C03	RED · 2 × 18	DATA — record twin size/profile within the blue-sized envelope.
W2 GND	C04 + C06	BLACK · 2 × 18	DATA — record twin size/profile within the blue-sized envelope.

14 AWG ferrules: three large red, one yellow, remaining device ends blue

Mean Well L/N, Waveshare COM/NO, detector A1/A2 and all Finder A1/A2: one 14 AWG wire in one blue ferrule.

14 AWG groups: two four-wire red, one three-wire red, one two-wire yellow and 16 single blue ends.

Kit chart: large red = 8 AWG label / 12 mm / 4.9 mm OD; yellow = 10 AWG label / 12 mm / 3.9 mm OD.

K4-14: one orange C13. K4-11: one black C14. The GPIO1/R1 branch is soldered locally at JS2; no shared ferrule at K4.

Finder 22.32 contact terminals are rated for 1 × 10 AWG or 2 × 12 AWG. K5-3 and K1-3 exceed that; use a rated neutral terminal block instead.

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

6 requested shared ferrules

Every wire end

Harness revision 13 — ferrule branches at existing devices. 27 IDs / 54 ends; six shared ferrules. Set the Mean Well supply to 5 V. After assembly, measure 5.00 V at the DC supply output with all relays closed, then confirm Home Assistant/RS485 communication, outlet sensing, a normal timed run, STOP and flow-loss shutdown. Calibrate with water before chemical service; see pump-calibration.md. Completed wiring does not replace these measurements. SINGLE = one conductor; SHARED = the stated number of conductors in one ferrule. All 120 V wiring stays 14 AWG. FITTED = as built; DATA = specifications to record; CHECK = end preparation. C12 is an integral resistor lead.

Wire	Insulation / conductor	End A · exact location	A · prepare this end	End B · exact location	B · prepare this end
A01	Black 14 AWG MTW, 600 V	BOX:L Inside metal box: hot Wago joins source-whip black to separate black A01 lead OUT	BOX Wago connection inside the metal box; follow the installed connector conductor range and strip length. No ferrule specified at this end.	F1:LINE F1 bottom LINE clamp; A01 incoming supply	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. C60 wire guidance: 14 mm strip and 22 lb-in for this current/voltage range, subject to installed markings. Kit blue barrel is 8 mm; check the metal-barrel engagement against the device instructions.
A02	Black 14 AWG MTW, 600 V	F1:LOAD F1 top LOAD clamp; one installed large-red ferrule with four black 14 AWG branches	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A02/A05/A06/A07, four black 14 AWG wires, tightened and connected as drawn. Wires: A02 + A05 + A06 + A07. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	PS1:L PS1 bottom RIGHT L, terminal 4	SINGLE 14 Single blue 14 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. Published wire strip 6 mm, torque 0.49 N·m; kit barrel is 8 mm.
A05	Black 14 AWG MTW, 600 V	F1:LOAD F1 top LOAD clamp; one installed large-red ferrule with four black 14 AWG branches	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A02/A05/A06/A07, four black 14 AWG wires, tightened and connected as drawn. Wires: A02 + A05 + A06 + A07. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	W2:COM W2 Slave / K3: bottom 2nd screw COM	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
A06	Black 14 AWG MTW, 600 V	F1:LOAD F1 top LOAD clamp; one installed large-red ferrule with four black 14 AWG branches	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A02/A05/A06/A07, four black 14 AWG wires, tightened and connected as drawn. Wires: A02 + A05 + A06 + A07. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K1:1 K1 Finder: top outer LEFT screw 1	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm.
A07	Black 14 AWG MTW, 600 V	F1:LOAD F1 top LOAD clamp; one installed large-red ferrule with four black 14 AWG branches	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A02/A05/A06/A07, four black 14 AWG wires, tightened and connected as drawn. Wires: A02 + A05 + A06 + A07. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	W1:COM W1 Master / K2: bottom 2nd screw COM	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
A08	White 14 AWG MTW, 600 V	BOX:N Inside metal box: neutral Wago joins source-whip neutral, outlet-whip neutral and white A08 lead OUT	BOX Wago connection inside the metal box; follow the installed connector conductor range and strip length. No ferrule specified at this end.	K5:3 K5 Finder: top outer RIGHT / neutral branch screw 3	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A08/A11/A13/A14, four white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A08 + A11 + A13 + A14. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.

Wire	Insulation / conductor	End A · exact location	A · prepare this end	End B · exact location	B · prepare this end
A11	 White 14 AWG MTW, 600 V	K5:3 K5 Finder: top outer RIGHT / neutral branch screw 3	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A08/A11/A13/A14, four white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A08 + A11 + A13 + A14. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K1:3 K1 Finder: top outer RIGHT / neutral branch screw 3	3 × 14 / ONE FITTED: one LARGE RED ferrule holding A11/A15/A16, three white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A11 + A15 + A16. ONE LARGE RED ferrule containing THREE 14 AWG conductors, at K1 terminal 3. Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.
A13	 White 14 AWG MTW, 600 V	K5:3 K5 Finder: top outer RIGHT / neutral branch screw 3	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A08/A11/A13/A14, four white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A08 + A11 + A13 + A14. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	PS1:N PS1 bottom LEFT N, terminal 3	SINGLE 14 Single blue 14 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. Published wire strip 6 mm, torque 0.49 N·m; kit barrel is 8 mm.
A14	 White 14 AWG MTW, 600 V	K5:3 K5 Finder: top outer RIGHT / neutral branch screw 3	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A08/A11/A13/A14, four white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A08 + A11 + A13 + A14. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K5:A2 K5 Finder: lower inner ledge screw A2	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm. A1/A2: one 14 AWG wire in one blue ferrule; no branch at this coil terminal.
A15	 White 14 AWG MTW, 600 V	K1:3 K1 Finder: top outer RIGHT / neutral branch screw 3	3 × 14 / ONE FITTED: one LARGE RED ferrule holding A11/A15/A16, three white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A11 + A15 + A16. ONE LARGE RED ferrule containing THREE 14 AWG conductors, at K1 terminal 3. Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K1:A2 K1 Finder: lower inner ledge screw A2	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm. A1/A2: one 14 AWG wire in one blue ferrule; no branch at this coil terminal.
A16	 White 14 AWG MTW, 600 V	K1:3 K1 Finder: top outer RIGHT / neutral branch screw 3	3 × 14 / ONE FITTED: one LARGE RED ferrule holding A11/A15/A16, three white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A11 + A15 + A16. ONE LARGE RED ferrule containing THREE 14 AWG conductors, at K1 terminal 3. Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K4:A2 K4 top inner A2 input clamp	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. K4 supports single ferrules 0.2–2.5 mm ² ; strip 8 mm, torque 0.6–0.8 N·m.
B01	 Red 14 AWG MTW, 600 V	W1:NO W1 Master / K2: bottom 1st / LEFTMOST screw NO	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.	K1:A1 K1 Finder: upper inner ledge screw A1	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm. A1/A2: one 14 AWG wire in one blue ferrule; no branch at this coil terminal.
B02	 Red 14 AWG MTW, 600 V	W2:NO W2 Slave / K3: bottom 1st / LEFTMOST screw NO	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.	K5:A1 K5 Finder: upper inner ledge screw A1	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm. A1/A2: one 14 AWG wire in one blue ferrule; no branch at this coil terminal.

Wire	Insulation / conductor	End A · exact location	A · prepare this end	End B · exact location	B · prepare this end
B03	Red 14 AWG MTW, 600 V	K1:2 K1 Finder: bottom outer LEFT screw 2	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm.	K5:1 K5 Finder: top outer LEFT screw 1	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm.
B04	Red 14 AWG MTW, 600 V	K5:2 K5 Finder: bottom outer LEFT screw 2	2 × 14 / ONE FITTED: one YELLOW ferrule holding B04/B05, two red 14 AWG wires, tightened and connected as drawn. Wires: B04 + B05. ONE YELLOW ferrule containing TWO 14 AWG conductors, at K5 terminal 2. Pittsburgh 70195 kit chart: 10 AWG label, 12 mm barrel, 3.9 mm tube OD. This records the kit ferrule used, not a separate twin-ferrule product.	BOX:SW Inside metal box: switched-hot Wago joins red B04 lead IN to outlet-whip black	BOX Wago connection inside the metal box; follow the installed connector conductor range and strip length. No ferrule specified at this end.
B05	Red 14 AWG MTW, 600 V	K5:2 K5 Finder: bottom outer LEFT screw 2	2 × 14 / ONE FITTED: one YELLOW ferrule holding B04/B05, two red 14 AWG wires, tightened and connected as drawn. Wires: B04 + B05. ONE YELLOW ferrule containing TWO 14 AWG conductors, at K5 terminal 2. Pittsburgh 70195 kit chart: 10 AWG label, 12 mm barrel, 3.9 mm tube OD. This records the kit ferrule used, not a separate twin-ferrule product.	K4:A1 K4 top outer A1 input clamp	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. K4 supports single ferrules 0.2–2.5 mm²; strip 8 mm, torque 0.6–0.8 N·m.
C01	Red 18 AWG MTW, 600 V	PS1:+V PS1 top LEFT +V, terminal 1	SINGLE 18 Single 18 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. Published wire strip 6 mm, torque 0.49 N·m; kit barrel is 8 mm.	W2:VCC W2 Slave / K3: top power RIGHT (+5 V) screw VCC	SHARED 18 DATA: record the two-18-AWG twin size/profile within the blue-sized envelope and the Waveshare clamp strip/torque specification. Wires: C01 + C03. ONE two-wire ferrule containing TWO 18 AWG conductors. Record its size, barrel length and crimp profile. Check retention of each wire and metal-barrel engagement at the clamp.
C03	Red 18 AWG MTW, 600 V	W2:VCC W2 Slave / K3: top power RIGHT (+5 V) screw VCC	SHARED 18 DATA: record the two-18-AWG twin size/profile within the blue-sized envelope and the Waveshare clamp strip/torque specification. Wires: C01 + C03. ONE two-wire ferrule containing TWO 18 AWG conductors. Record its size, barrel length and crimp profile. Check retention of each wire and metal-barrel engagement at the clamp.	W1:VCC W1 Master / K2: top power RIGHT (+5 V) screw VCC	SINGLE 18 SINGLE 18 AWG ferrule. Kit reference: SMALL red collar, 8 mm barrel; kit chart metal-tube OD 1.7 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
C04	Black 18 AWG MTW, 600 V	PS1:-V PS1 top RIGHT −V, terminal 2	SINGLE 18 Single 18 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. Published wire strip 6 mm, torque 0.49 N·m; kit barrel is 8 mm.	W2:GND W2 Slave / K3: top power LEFT screw GND	SHARED 18 DATA: record the two-18-AWG twin size/profile within the blue-sized envelope and the Waveshare clamp strip/torque specification. Wires: C04 + C06. ONE two-wire ferrule containing TWO 18 AWG conductors. Record its size, barrel length and crimp profile. Check retention of each wire and metal-barrel engagement at the clamp.
C06	Black 18 AWG MTW, 600 V	W2:GND W2 Slave / K3: top power LEFT screw GND	SHARED 18 DATA: record the two-18-AWG twin size/profile within the blue-sized envelope and the Waveshare clamp strip/torque specification. Wires: C04 + C06. ONE two-wire ferrule containing TWO 18 AWG conductors. Record its size, barrel length and crimp profile. Check retention of each wire and metal-barrel engagement at the clamp.	W1:GND W1 Master / K2: top power LEFT screw GND	SINGLE 18 SINGLE 18 AWG ferrule. Kit reference: SMALL red collar, 8 mm barrel; kit chart metal-tube OD 1.7 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.

Wire	Insulation / conductor	End A · exact location	A · prepare this end	End B · exact location	B · prepare this end
C07	Blue One conductor of intact Cat6A pair	W1:A+ W1 Master / K2: bottom 4th / RIGHTMOST screw A+	CABLE END Identify Cat6A gauge/construction and clamp specification. Solid: bare. Stranded: only an accepted matching ferrule. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.	W2:A+ W2 Slave / K3: bottom 4th / RIGHTMOST screw A+	CABLE END Identify Cat6A gauge/construction and clamp specification. Solid: bare. Stranded: only an accepted matching ferrule. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
C08	White/blue stripe Mate of same intact Cat6A pair	W1:B- W1 Master / K2: bottom 3rd screw B-	CABLE END Identify Cat6A gauge/construction and clamp specification. Solid: bare. Stranded: only an accepted matching ferrule. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.	W2:B- W2 Slave / K3: bottom 3rd screw B-	CABLE END Identify Cat6A gauge/construction and clamp specification. Solid: bare. Stranded: only an accepted matching ferrule. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
C09	Red Installed telephone-cord conductor; gauge not recorded	JM1 JM1: Master J5-1 GPIO1 YELLOW factory lead to RED telephone-cord conductor C09; photographed splice	SOLDER Mechanically engaged solder joint, individually sleeved and strain-relieved; no ferrule.	RJ11:FLOW Installed RJ11 coupling at enclosure bottom: GPIO1 contact of the connected flow-switch pair; functional label, not a contact number	RJ11 CONTACT Existing telephone-cord RJ11 contact. Retain the installed contact mapping; this drawing assigns no numbered RJ11 pinout.
C10	Green Installed telephone-cord conductor; gauge not recorded	JM2 JM2: Master J5-4 GND BLACK factory lead to GREEN telephone-cord conductor C10; photographed splice	SOLDER Mechanically engaged solder joint, individually sleeved and strain-relieved; no ferrule.	RJ11:RETURN Installed RJ11 coupling at enclosure bottom: GND contact of the connected flow-switch pair; functional label, not a contact number	RJ11 CONTACT Existing telephone-cord RJ11 contact. Retain the installed contact mapping; this drawing assigns no numbered RJ11 pinout.
C12	Metal lead Integral R1 lead	R1:SENSE Integral R1 sense-side lead; C12 is not cut hookup wire	R1 LEAD Integral resistor lead C12; sleeve the exposed metal and its splice. No extra hookup wire or ferrule at this end.	JS2 JS2: ONE local three-conductor splice joins YELLOW GPIO1 factory lead, R1 lead C12 and ORANGE C13 to detector terminal 14	SOLDER Three-conductor solder splice: YELLOW GPIO1 factory lead + R1 lead C12 + ORANGE C13. Individually insulate this node from the RED 3V3 splice and support the resistor assembly.
C13	Orange 18 AWG MTW, 600 V	JS2 JS2: ONE local three-conductor splice joins YELLOW GPIO1 factory lead, R1 lead C12 and ORANGE C13 to detector terminal 14	SOLDER Three-conductor solder splice: YELLOW GPIO1 factory lead + R1 lead C12 + ORANGE C13. Individually insulate this node from the RED 3V3 splice and support the resistor assembly.	K4:14 K4 bottom middle 14 normally open	SINGLE 18 SINGLE 18 AWG ferrule. Kit reference: SMALL red collar, 8 mm barrel; kit chart metal-tube OD 1.7 mm. Check device-specific note. K4 supports single ferrules 0.2–2.5 mm²; strip 8 mm, torque 0.6–0.8 N·m.
C14	Black 18 AWG MTW, 600 V	JS3 JS3: Slave J5-4 GND BLACK factory lead joined to BLACK C14 for detector terminal 11	SOLDER Mechanically engaged solder joint, individually sleeved and strain-relieved; no ferrule.	K4:11 K4 bottom inner 11 common	SINGLE 18 SINGLE 18 AWG ferrule. Kit reference: SMALL red collar, 8 mm barrel; kit chart metal-tube OD 1.7 mm. Check device-specific note. K4 supports single ferrules 0.2–2.5 mm²; strip 8 mm, torque 0.6–0.8 N·m.